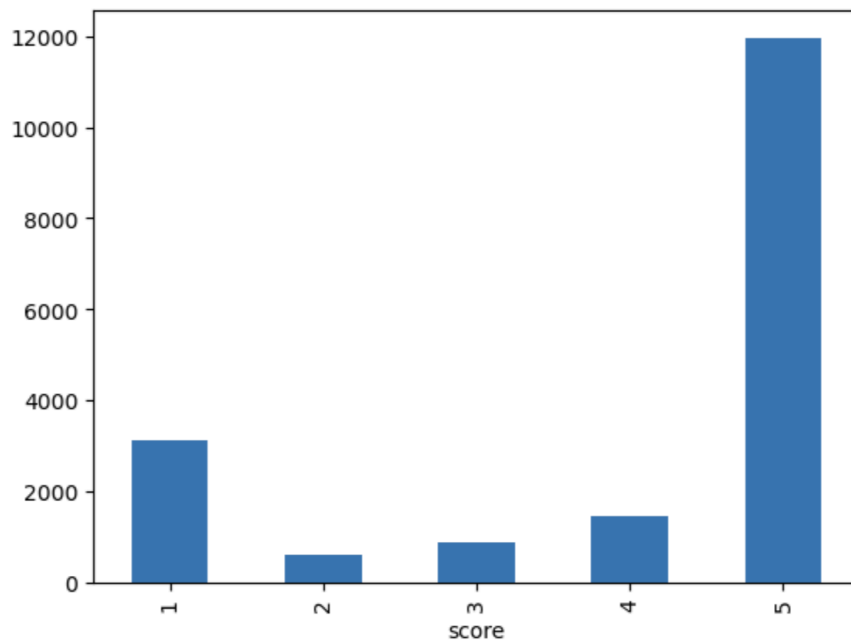
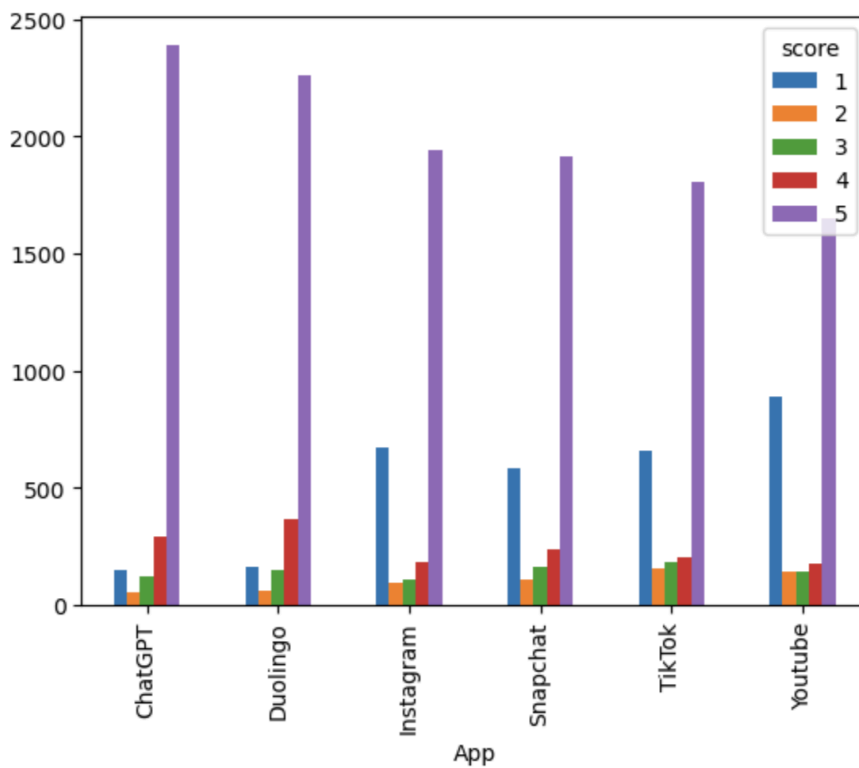


1. Rating Distribution

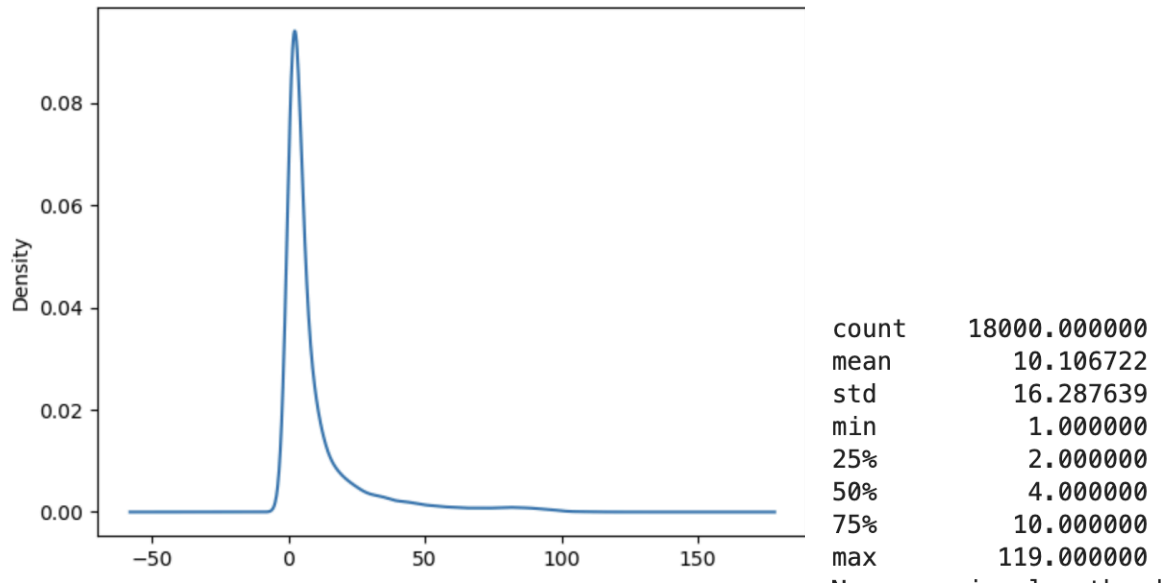


The distribution of ratings for six apps in total is skewed with the highest peak at score 5.



The distributions of ratings for each app are very similar. Instagram, Snapchat, TikTok and Youtube have more score 1 compared to the other two apps.

2. Text Length



The text length without spaces has the most common length of 4.

	count	mean	median
score			
1	3104	21.237113	12.0
2	611	21.525368	12.0
3	866	15.702079	7.0
4	1458	10.781893	4.0
5	11961	6.147563	3.0

Low scores tend to have longer text, and a high score tend to have a short text, which is a good thing because we are more likely to extract commercial value from low score but longer text.

3. Timestamp Coverage

The earliest date is 6/27, and the latest date is 7/6

4. Missing Fields

```
App                0
reviewId           0
userImage          0
content            0
score              0
thumbsUpCount      0
reviewCreatedVersion 3450
date               0
review_length      0
..               ..
```

The original data extracted has null values for columns: replyContent, repliedAt, appVersion. Since these three columns are not helpful for the later sentiment analysis, I dropped them in the "df_clean" for EDA. From the result above, we still have NAs for the column "reviewCreatedVersion".

5. Duplicate IDs

We do not have duplicated IDs in the dataset, which is good.

6. Repeated Review Text

```
# repeated review text
duplicate_reviews = df_clean['content'].duplicated().sum()
duplicate_reviews
```

```
np.int64(5002)
```

```
df_clean['content'].value_counts().head(10)
```

```
content
good      961
nice      402
Good      248
very good  194
super     130
best      120
good app  118
nice app  113
good 🍌    97
ok        95
..       ..
```

We do have repeated text in the dataset, and the word "good" has the highest frequency. We have a total of 5002 repeated text, which takes about 1/3 of the entire dataset, so this is a concern.

7. Language Issue

I only scraped English reviews, so we are good to analyze reviews for English only reviews later.

8. Low Signal Reviews

I have three criteria for checking the usefulness of the reviews:

a. Short Reviews

We have 9918 short reviews in total, which is a big number given our sample size of 18000.

b. Emoji Only Content

We have 448 emoji only reviews, which contributes nothing to the commercial value.

c. No Actual Word Included

We have 851 reviews that contain no characters, which includes the emoji only content as above.

For this part, I think half of the dataset contains no meaningful information, which may have a big impact on our later sentiment analysis.